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VLSI Design Laboratory Workbook

CODE: PCCECE593

Sougata Chandra

ECE – 3A – 07

ELECTRONICS AND COMMUNICATION ENGINEERING DEPARTMENT



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PAPER NAME: **VLSI LABORATORY**

PAPER CODE: **PCCEC593**

Signature of the Lab-in-Charge

EXPERIMENT LIST

EXPERIMENT NO.	DIGITAL ELECTRONICS EXPERIMENT NAME
1	Step-by-Step Design and Simulation of a CMOS Inverter using Cadence® Virtuoso©
2	Design and Simulation of Basic Gates using Cadence® Virtuoso©
3	Design & Building of Logic Symbol Library of Basic Gates
4	Design and Simulation of Half and Full Adder using CMOS180 Technology
5	Design and Simulation of Half and Full Subtractor using CMOS180 Technology
6	Design and Simulation of 4-bit Full Adder using CMOS180 Technology
7	Layout Design using Cadence Layout XL using CMOS180 Technology
8.	Design and simulation of Combinational Logic Circuits using CMOS180 Technology
8.1	LAYOUT DESIGN AND SIMULATION OF CMOS INVERTER
8.2	LAYOUT DESIGN AND SIMULATION OF NOR GATE
8.3	LAYOUT DESIGN AND SIMULATION OF NAND GATE
8.4	LAYOUT DESIGN AND SIMULATION OF AND GATE & OR GATE
8.5	LAYOUT DESIGN AND SIMULATION OF EXCLUSIVE GATES
8.6	LAYOUT DESIGN AND SIMULATION OF HALF-ADDER
8.7	LAYOUT DESIGN AND SIMULATION OF FULL-ADDER

INDEX

EXP. NO.	Sub-Exp. No.	NAME OF EXPERIMENT	SIGNATURE	REMARKS
1	1	Step-by-Step Design and Simulation of a CMOS Inverter using Cadence® Virtuoso©		
2	1	Design and Simulation of Basic Gates using Cadence® Virtuoso© (NAND GATE)		
2	2	Design and Simulation of Basic Gates using Cadence® Virtuoso© (NOR GATE)		

Exp. No.: 1	**Step-by-Step Design and Simulation of a CMOS Inverter using Cadence® Virtuoso©**

AIM:

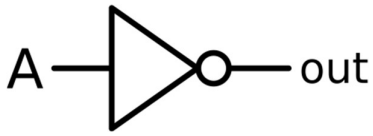
To study the simulation and implementation procedures of the design of logic gates using CMOS Technology with UMC 180nm Library **using Cadence® Virtuoso©**

NOT GATE:

The Inverter performs a basic logic gate function called Inversion or Complementation. The purpose of an inverter is to change one logic level to opposite level. When a high level is applied to an inverter, the low level will appear at the output and vice versa.

$$Y = \sim a$$

Not Gate:

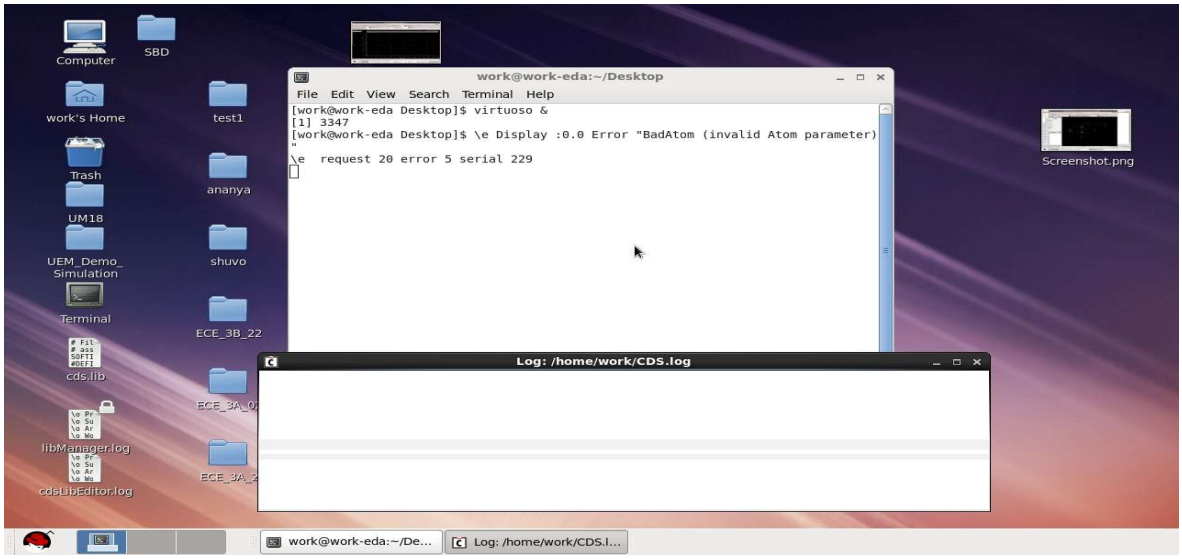


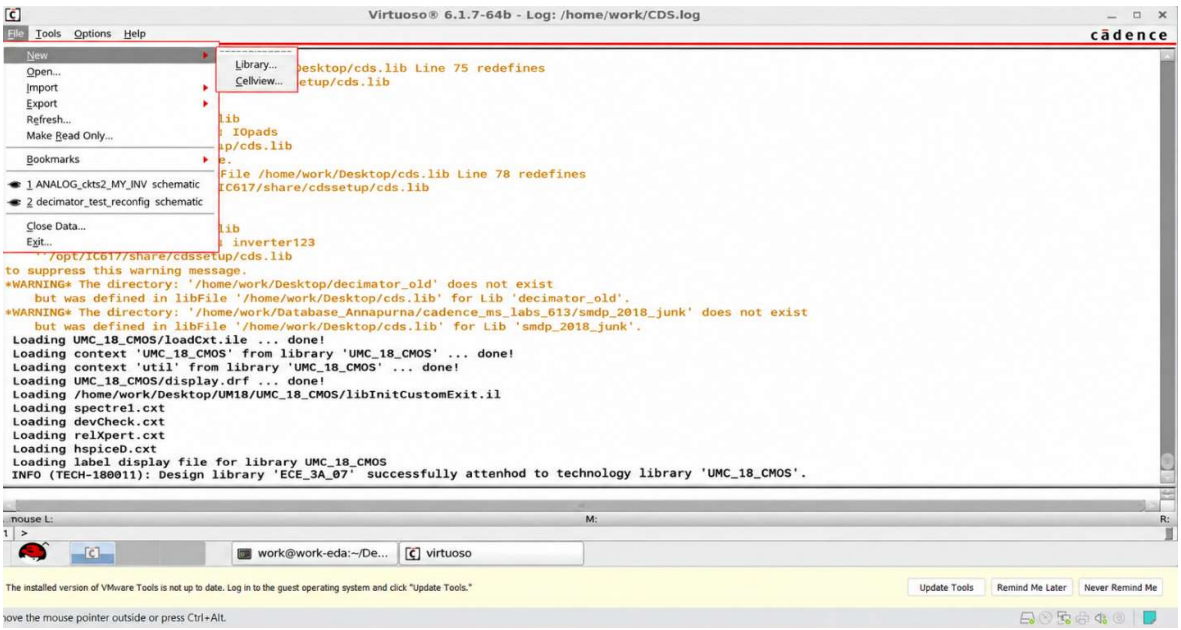
Truth table:

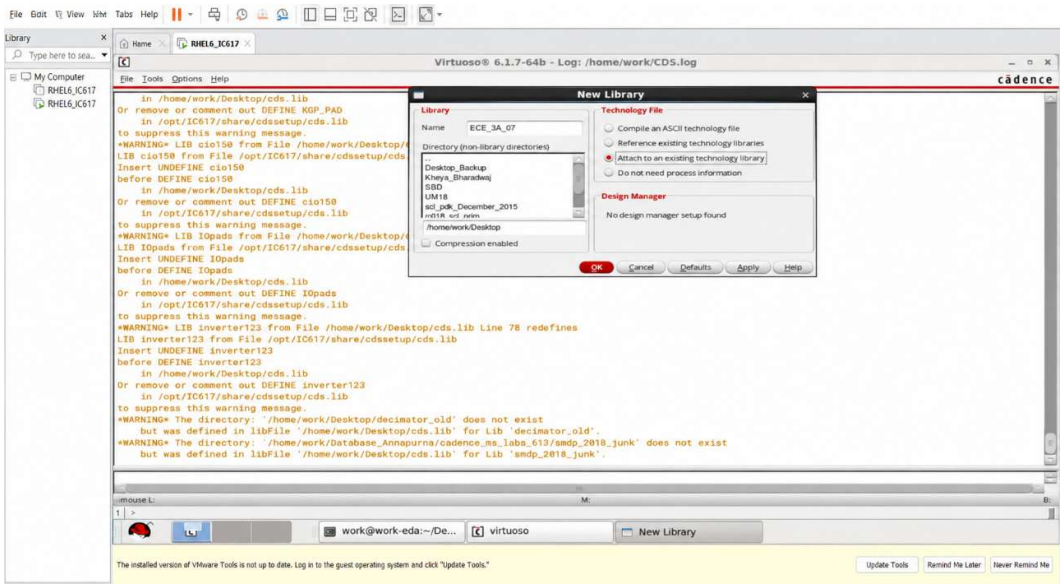
A	Y
0	1
1	0

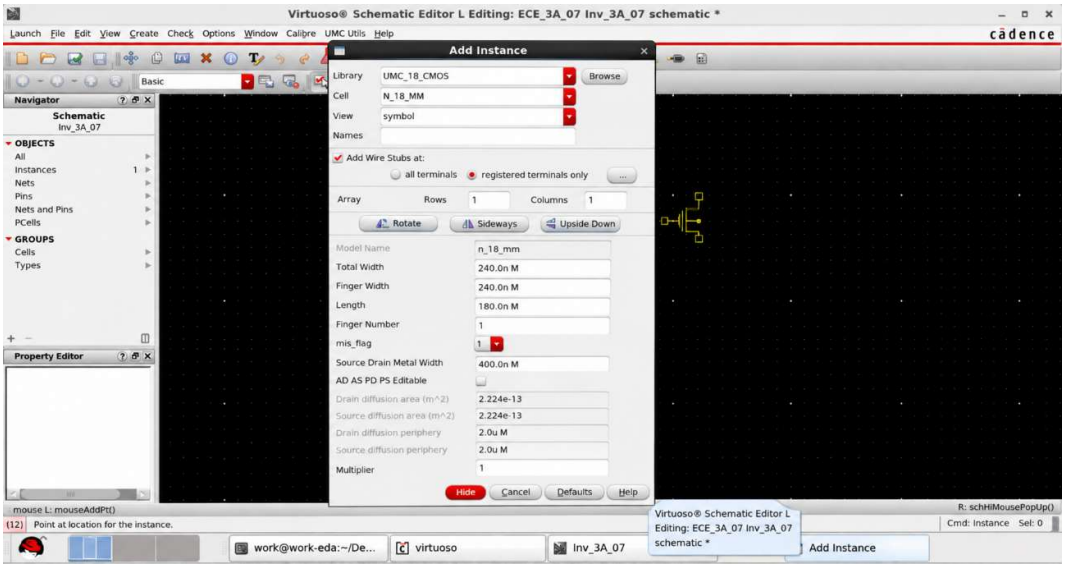
DESIGN AND SIMULATION SET-UP

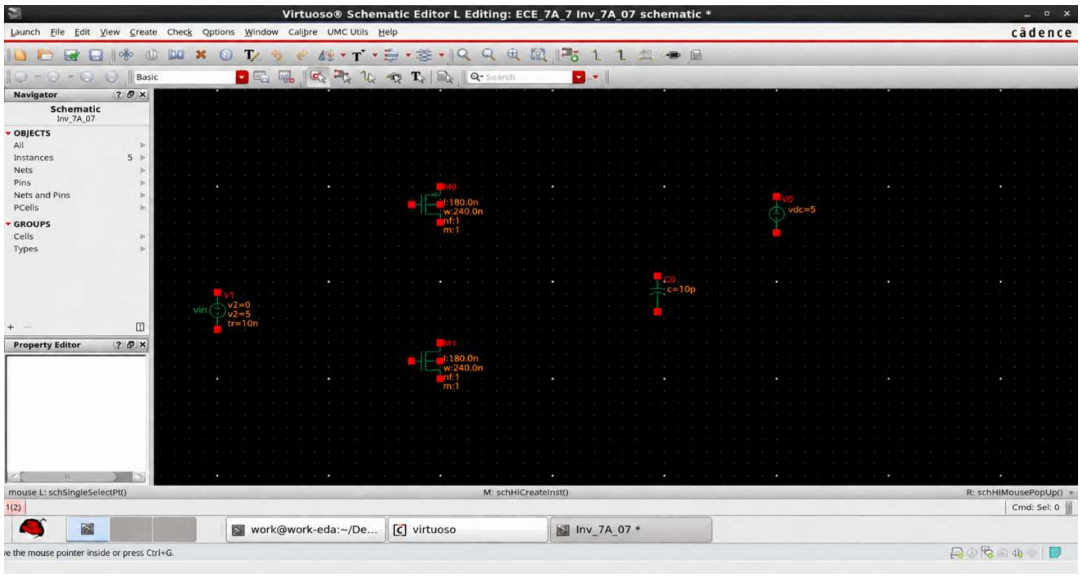
a) Design

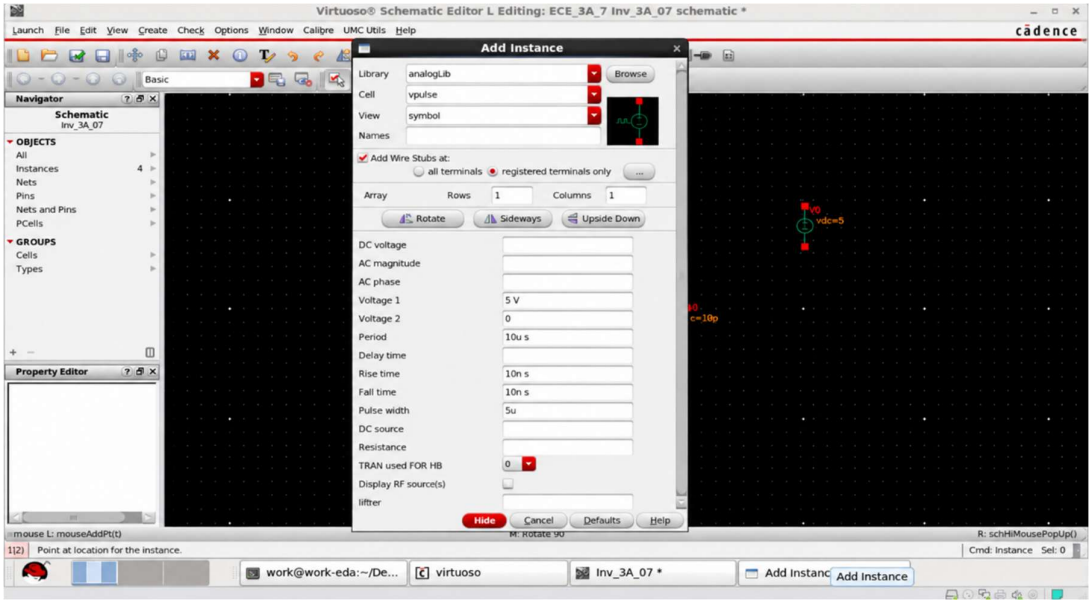


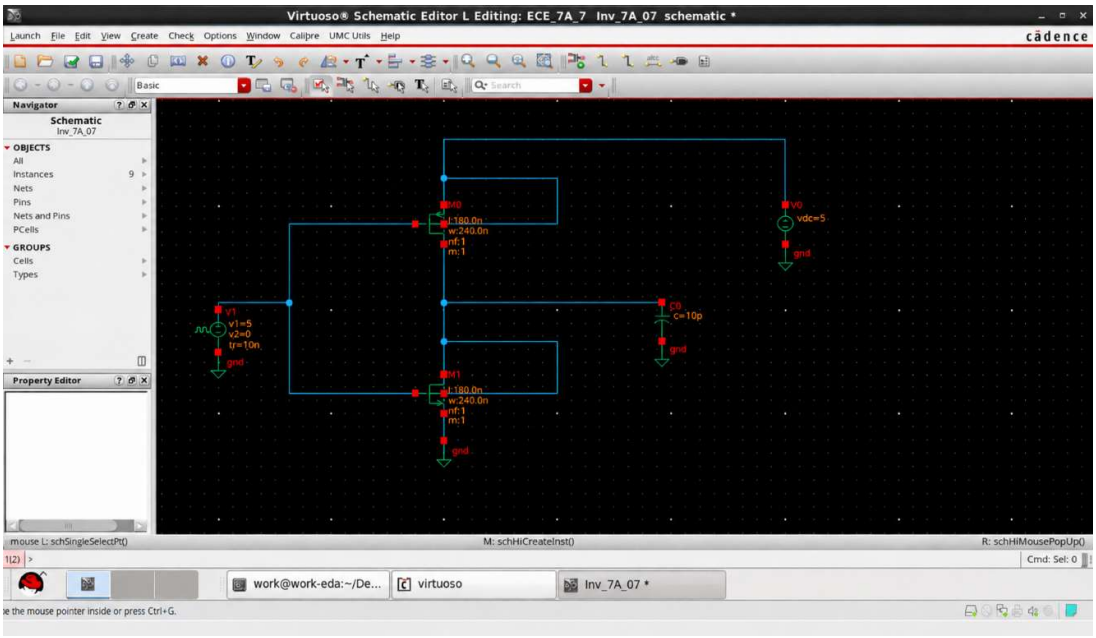



Virtuoso® 6.1.7-64b - Log: /home/work/CDS.log
New Library
Library
Name
ECE_3A_07
Technology File
Compile an ASCII technology file
Reference existing technology libraries
Attach to an existing technology library
Do not need process information
Design Manager
No design manager setup found
Desktop_Backup
Kheya_Bharadwaj
SBD
UM18
scl_pdk_December_2015
/home/work/Desktop
Compression enabled
OK
Cancel
Defaults
Apply
Help
WARNING LIB inverter123 from File /home/work/Desktop/cds.lib Line 78 redefines
WARNING The directory: '/home/work/Desktop/decimator_old' does not exist
but was defined in libFile '/home/work/Desktop/cds.lib' for Lib 'decimator_old'.
WARNING The directory: '/home/work/Database_Annapurna/cadence_ms_labs_613/smdp_2018_junk' does not exist
but was defined in libFile '/home/work/Desktop/cds.lib' for Lib 'smdp_2018_junk'.
work@work-eda:~/De...
virtuoso
New Library
The installed version of VMware Tools is not up to date. Log in to the guest operating system and click "Update Tools."
Update Tools
Remind Me Later
Never Remind Me

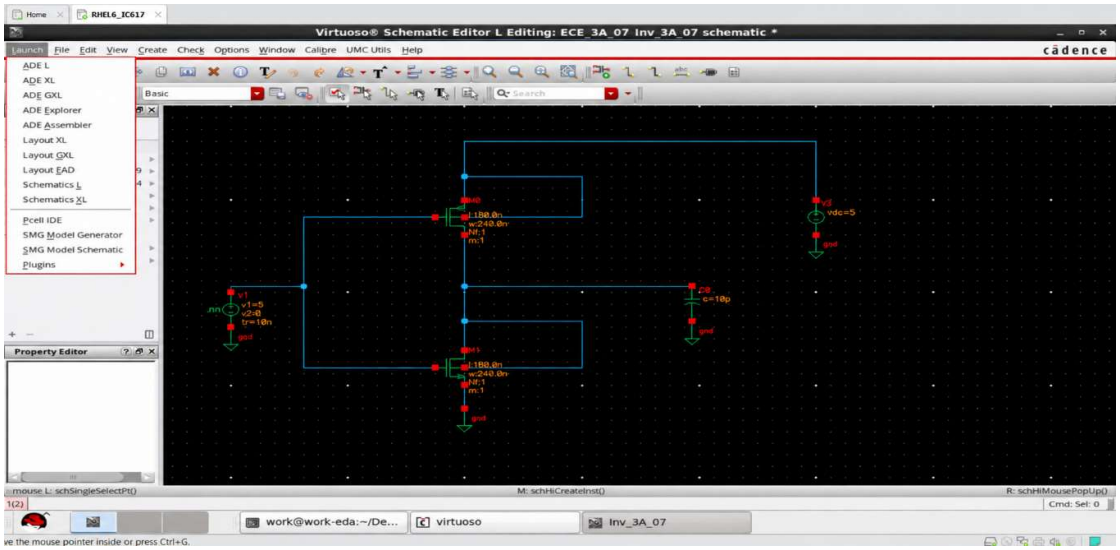

Virtuoso® Schematic Editor L Editing: ECE_3A_07 Inv_3A_07 schematic *
Add Instance
Library
UMC_18_CMOS
Browse
Cell
N_18_MM
View
symbol
Names
Add Wire Stubs at:
all terminals
registered terminals only
Array
Rows
1
Columns
1
Rotate
Sideways
Upside Down
Model Name
n_18_mm
Total Width
240.0n M
Finger Width
240.0n M
Length
180.0n M
Finger Number
1
mis_flag
Source Drain Metal Width
400.0n M
AD AS PD PS Editable
Drain diffusion area (m^2)
2.224e-13
Source diffusion area (m^2)
2.224e-13
Drain diffusion periphery
2.0u M
Source diffusion periphery
2.0u M
Multiplier
1
Hide
Cancel
Defaults
Help
mouse L: mouseAddPt()
Point at location for the instance.
R: schHiMousePopUp()
Cmd: Instance Sel: 0
work@work-eda:~/De...
virtuoso
Inv_3A_07
Add Instance
Virtuoso® Schematic Editor L Editing: ECE_3A_07 Inv_3A_07 schematic *

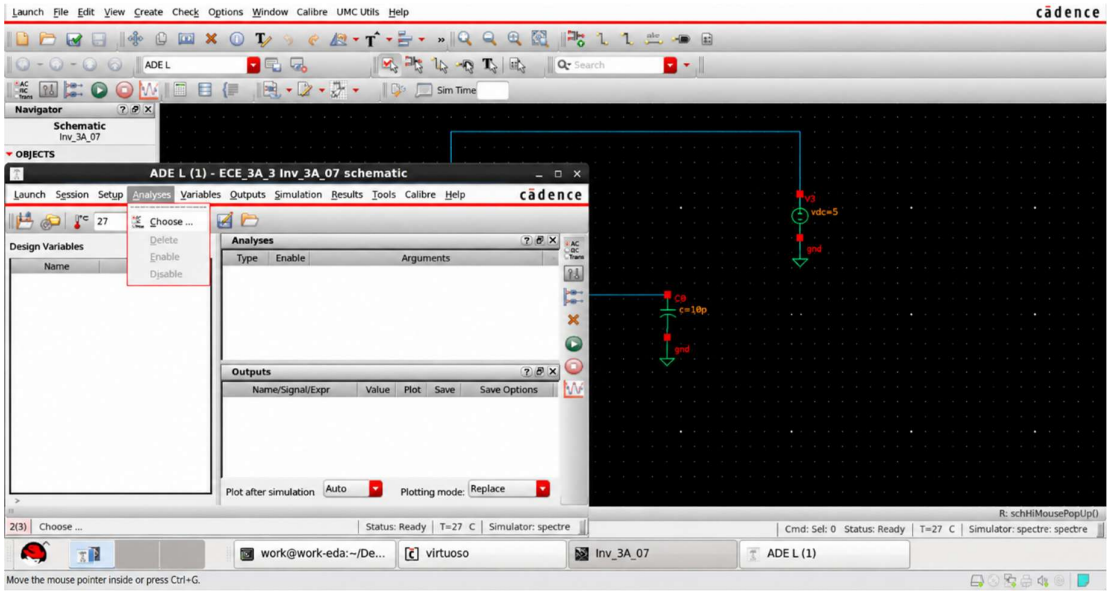

Virtuoso® Schematic Editor L Editing: ECE_7A_7 Inv_7A_07 schematic *
vdc=5
c=10p
vin
v2=0
v2=5
tr=10n
l:180.0n
w:240.0n
nf:1
m:1
mouse L: schSingleSelectPt()
M: schHiCreateInst()
R: schHiMousePopUp()
Cmd: Sel: 0
work@work-eda:~/De...
virtuoso
Inv_7A_07 *

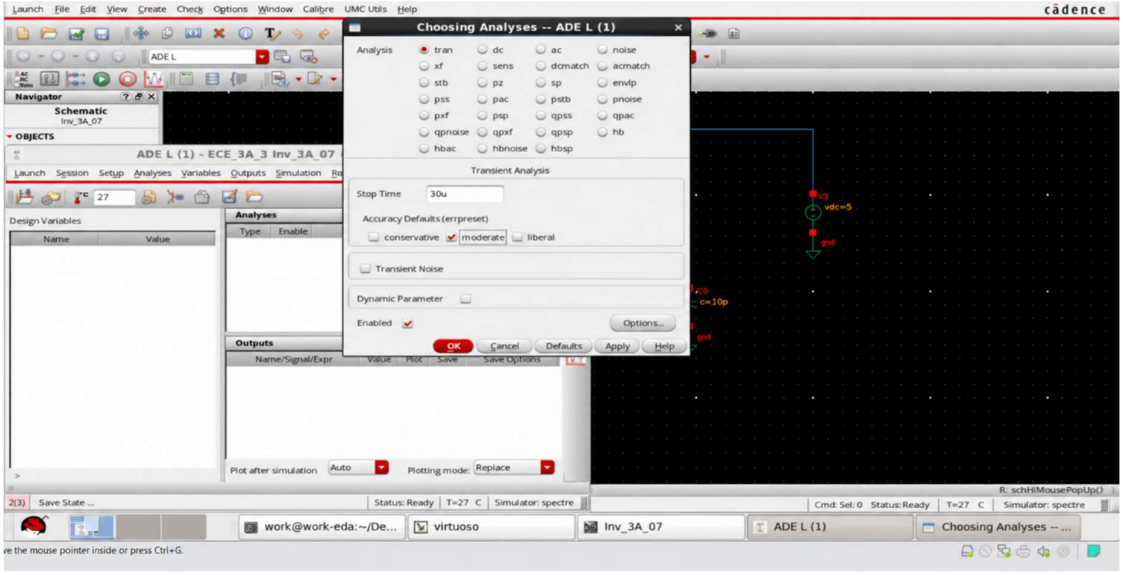




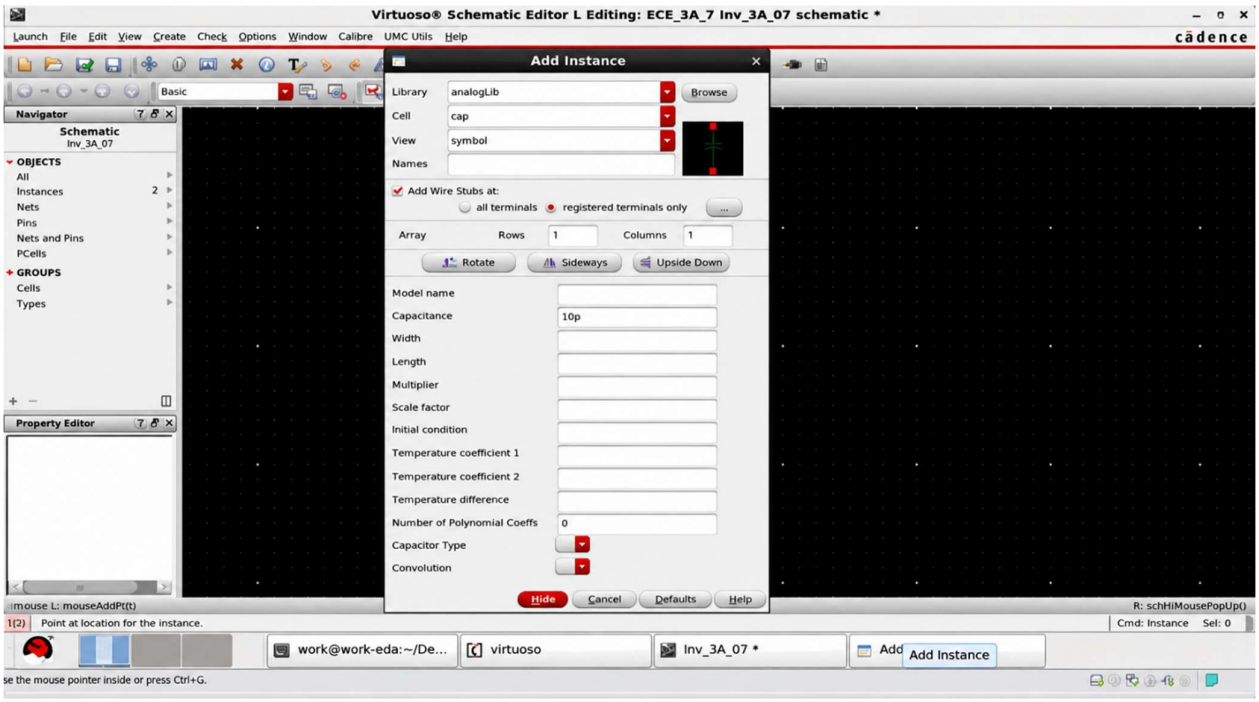
b) ADEL Simulation Setup


Virtuoso® Schematic Editor L Editing: ECE_3A_07 Inv_3A_07 schematic *
cadence
Launch File Edit View Create Check Options Window Calibre UMC Utils Help
ADE L
ADE XL
ADE GXL
ADE Explorer
ADE Assembler
Layout XL
Layout GXL
Layout EAD
Schematics L
Schematics XL
Pcell IDE
SMG Model Generator
SMG Model Schematic
Plugins
Property Editor
work@work-eda:~/De...
virtuoso
Inv_3A_07


Launch File Edit View Create Check Options Window Calibre UMC Utils Help
cadence
ADE L (1) - ECE_3A_3 Inv_3A_07 schematic
Launch Session Setup Analyses Variables Outputs Simulation Results Tools Calibre Help
Choose ...
Delete
Enable
Disable
Design Variables
Name
Analyses
Type Enable Arguments
Outputs
Name/Signal/Expr Value Plot Save Save Options
Plot after simulation Auto
Plotting mode: Replace
Navigator
Schematic
Inv_3A_07
OBJECTS
Sim Time
Status: Ready
Simulator: spectre
work@work-eda:~/De...
virtuoso
Inv_3A_07
ADE L (1)
Move the mouse pointer inside or press Ctrl+G.


Launch File Edit View Create Check Options Window Calibre UMC Utils Help
cadence
Choosing Analyses -- ADE L (1)
Analysis
tran dc ac noise
xf sens dcmatch acmatch
stb pz sp envlp
pss pac pstb pnoise
pxf psp qpss qpac
qpnoise qpxf qpsp hb
hbac hbnoise hbsp
Transient Analysis
Stop Time 30u
Accuracy Defaults (errpreset)
conservative moderate liberal
Transient Noise
Dynamic Parameter
Enabled
Options...
OK Cancel Defaults Apply Help
ADE L (1) - ECE_3A_3 Inv_3A_07
Design Variables
Name Value
Analyses
Outputs
Plot after simulation Auto
Plotting mode: Replace
Navigator
Schematic
Inv_3A_07
OBJECTS
Status: Ready
Simulator: spectre
work@work-eda:~/De...
virtuoso
Inv_3A_07
ADE L (1)
Choosing Analyses -- ...
ve the mouse pointer inside or press Ctrl+G.

c) Transient Simulation Result:



d) Conclusion:

In this experiment, we have learnt the design of a CMOS inverter using the UMC 180 nm technology library (UMC_18_CMOS PDK) in a professional SPICE environment (Cadence Virtuoso). Subsequently, the CMOS circuit has been validated using transient simulation in the ADE-L environment. From this experiment we learnt how a PMOS and an NMOS transistor are connected in a complementary manner to realize the inverter, how the devices are selected from the foundry library and placed in the schematic editor, and how the schematic is checked and saved before creating a symbol for it. We also learnt to build a test bench with the DC supply and a pulse input, to set up the transient analysis in ADE-L, and to plot the input and output waveforms together.

The simulated output verifies the inverting action of the circuit. The output is high when the input is low and low when the input is high, and the transitions follow the applied input without any loss in the logic levels. This confirms the correctness of both the schematic and the simulation setup, and establishes the basic design flow in Cadence Virtuoso that will be used in the subsequent experiments.

Signature of the Lab-in-Charge

Exp. No.: 2	Design and Simulation of Basic Gates using Cadence® Virtuoso©

AIM:

To study the simulation and implementation procedures **using Cadence® Virtuoso©**

THEORY:

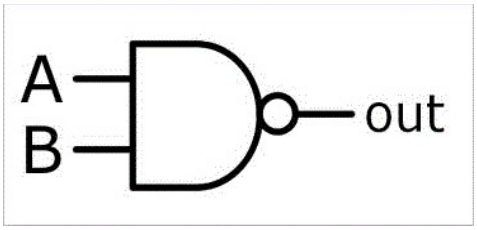
NAND GATE:

The term NAND is derived from the complement of AND. It implies the AND junction with an inverted output. The operation of NAND gate is such that the outputis low only when all its inputs are high and when any one of the inputs is low the output is high.

$$Y = \sim(a \,\&\, b)$$

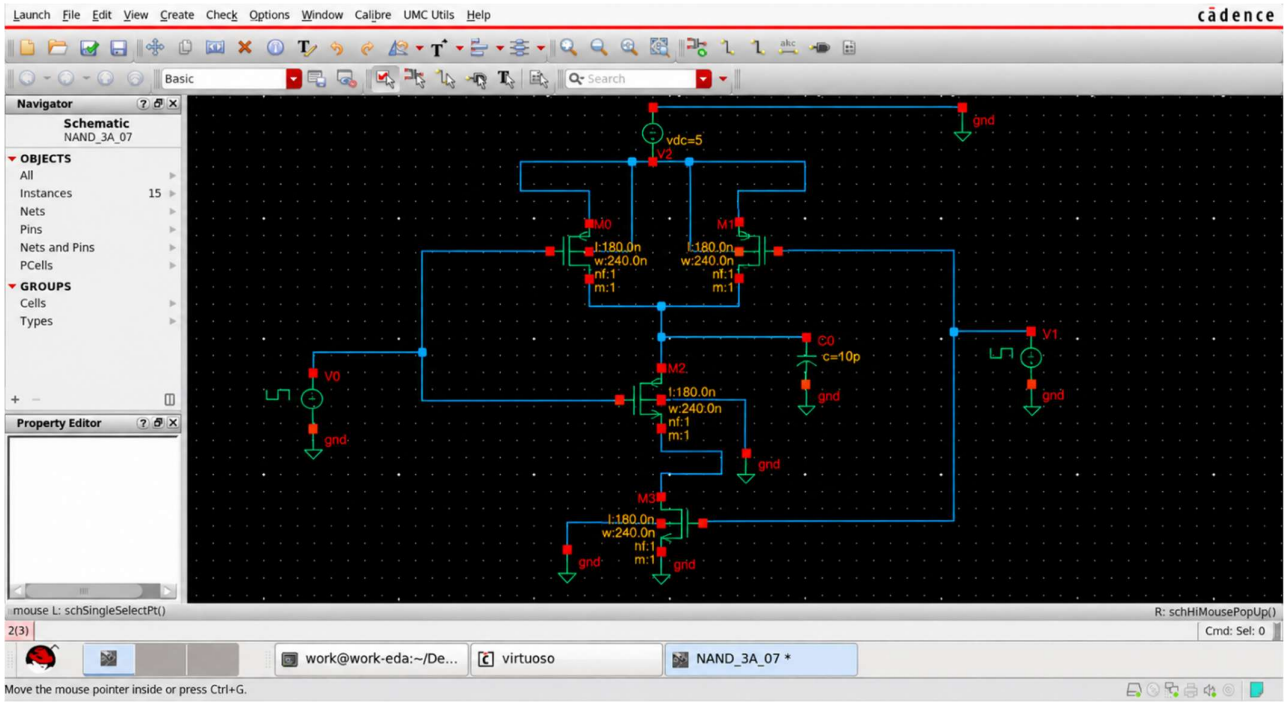
Truth table:

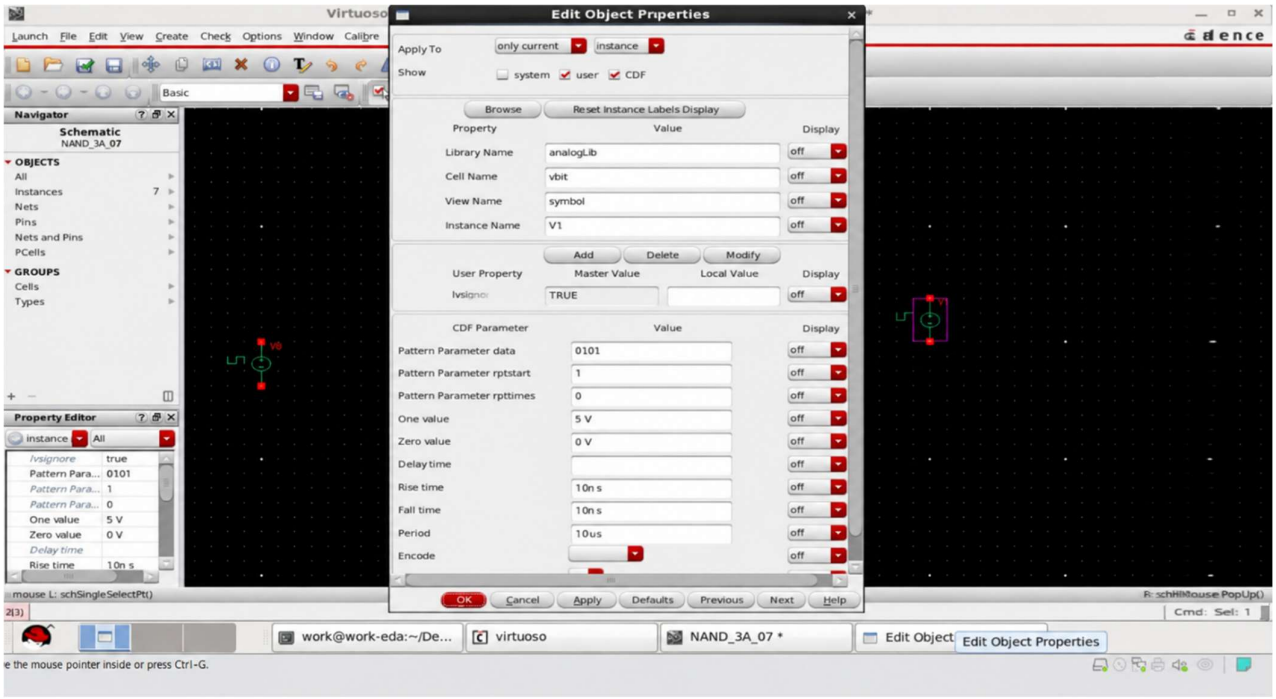
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



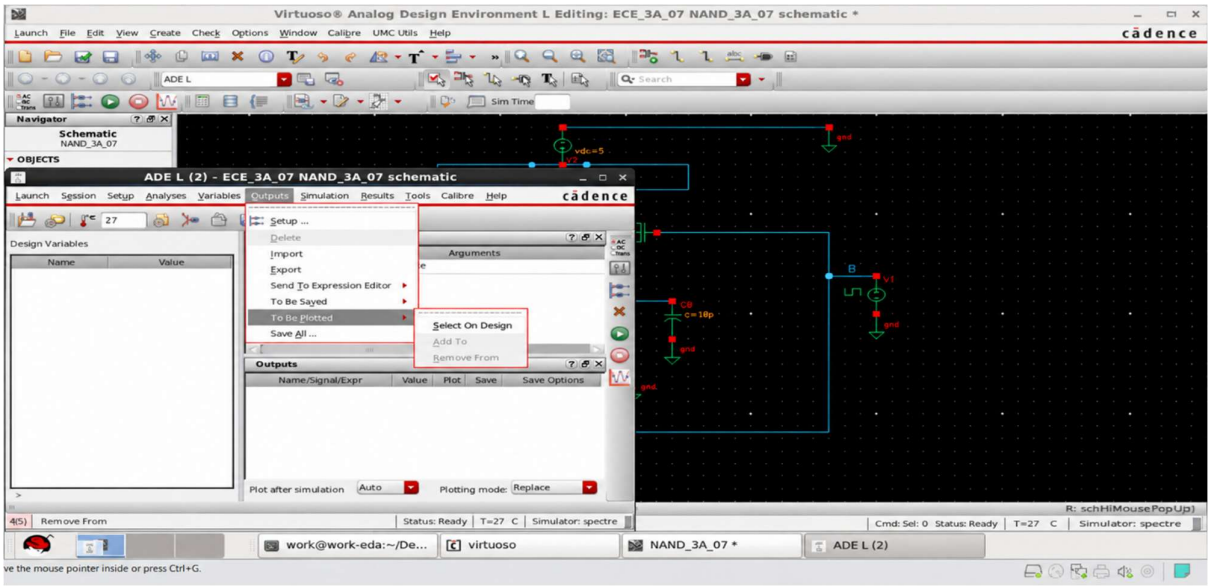
DESIGN AND SIMULATION SET-UP:

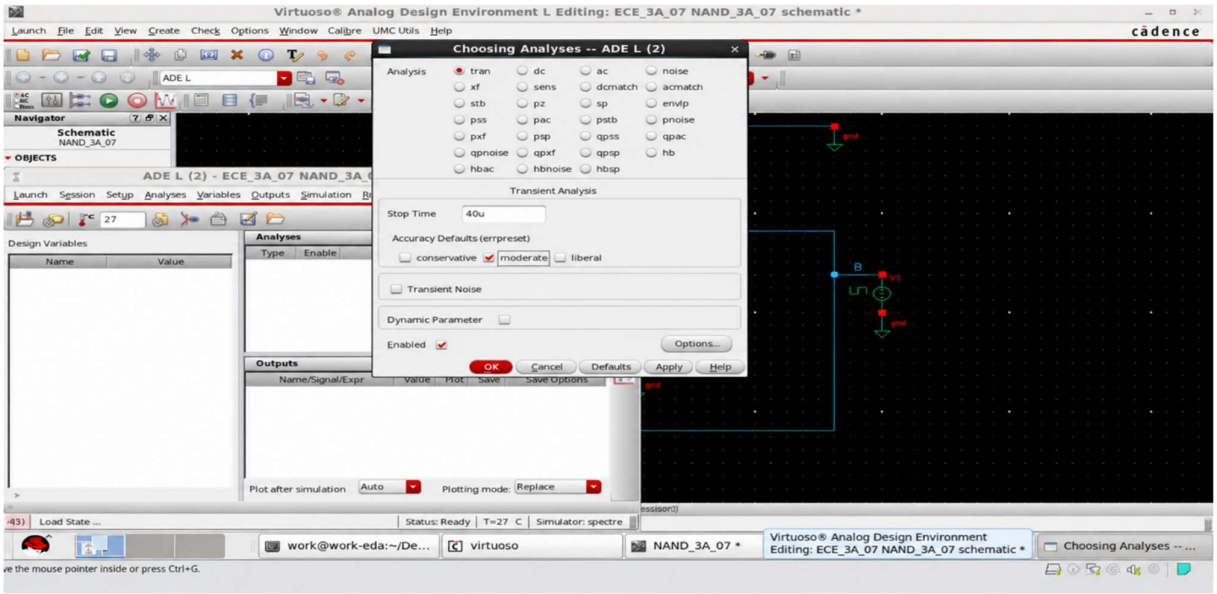
a) Design





b) ADEL Simulation Setup





c) Transient Simulation Result



d) Conclusion

In this experiment, we have learnt the design of a CMOS NAND gate using the UMC 180 nm technology library (UMC_18_CMOS PDK) in a professional SPICE environment (Cadence Virtuoso). Subsequently, the CMOS circuit has been validated using transient simulation in the ADE-L environment. From this experiment we learnt how PMOS and NMOS transistors are connected in complementary pull-up and pull-down networks to realize the NAND logic function, how the devices are selected from the foundry library and placed in the schematic editor, and how the schematic is checked and saved before creating a symbol for it. We also learnt to build a test bench with the DC supply and multiple pulse inputs, to configure the transient analysis in ADE-L, and to plot the input and output waveforms together for verification.

The simulated output verifies the NAND operation of the circuit. The output remains high for all input combinations except when both inputs are high, for which the output becomes low. The output transitions follow the applied input signals correctly and maintain the expected logic levels throughout the simulation. This confirms the correctness of both the CMOS NAND schematic and the simulation setup, and establishes the basic design methodology for implementing multi-input CMOS logic gates in Cadence Virtuoso.

Signature of the Lab-in-Charge

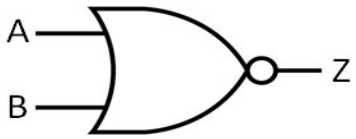
NOR GATE:

The term NOR is derived from the complement of OR. It implies the OR junction with an inverted output. The operation of NOR gate is such that the output is high only when all its inputs are low and when any one of the inputs is high the output is low.

$$Y = \sim(a \mid b)$$

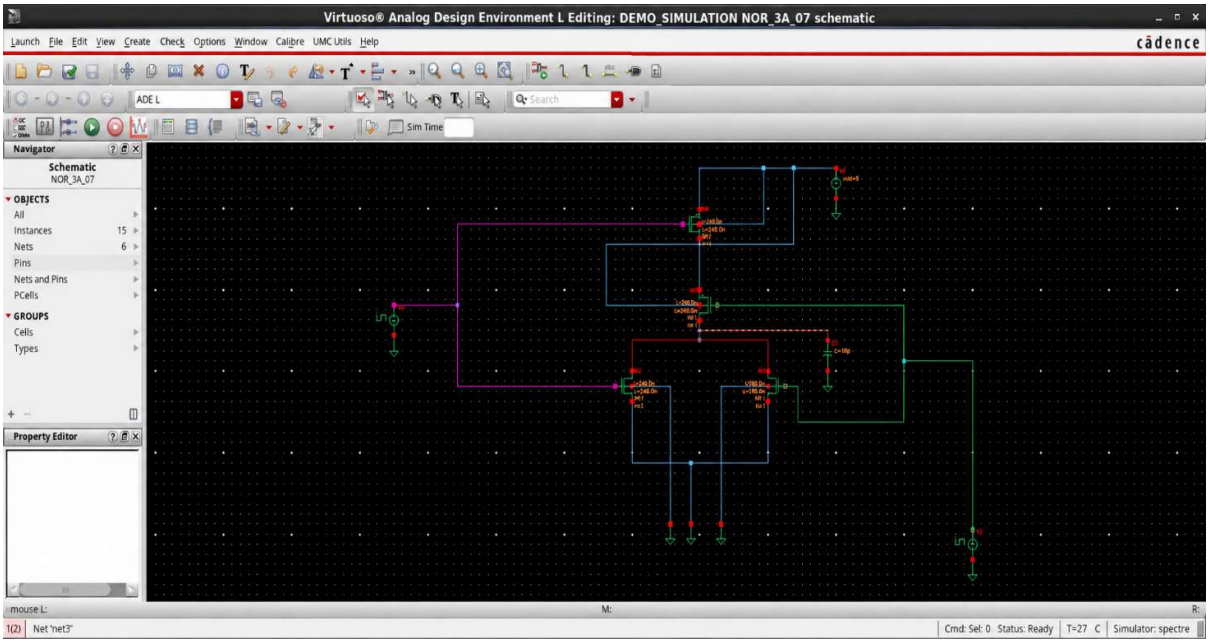
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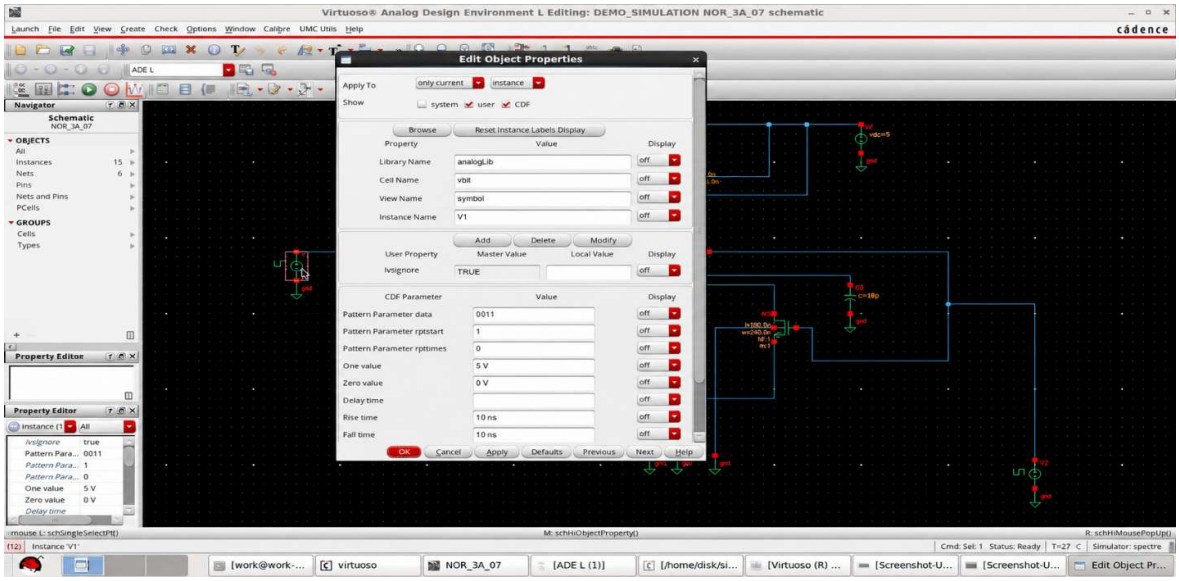
A	B	Y

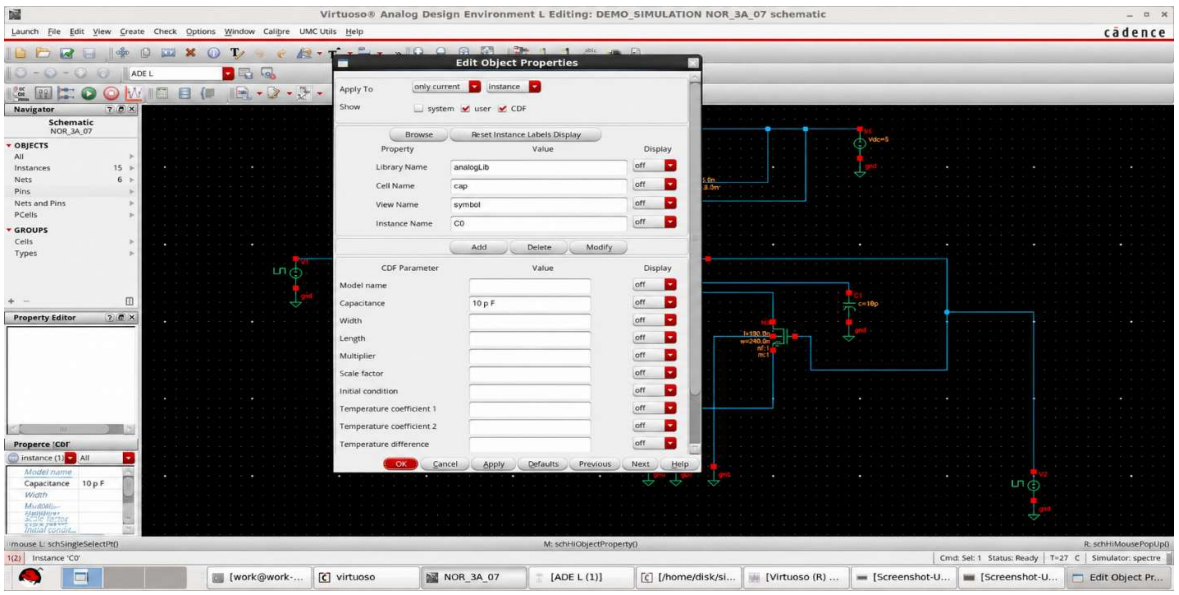


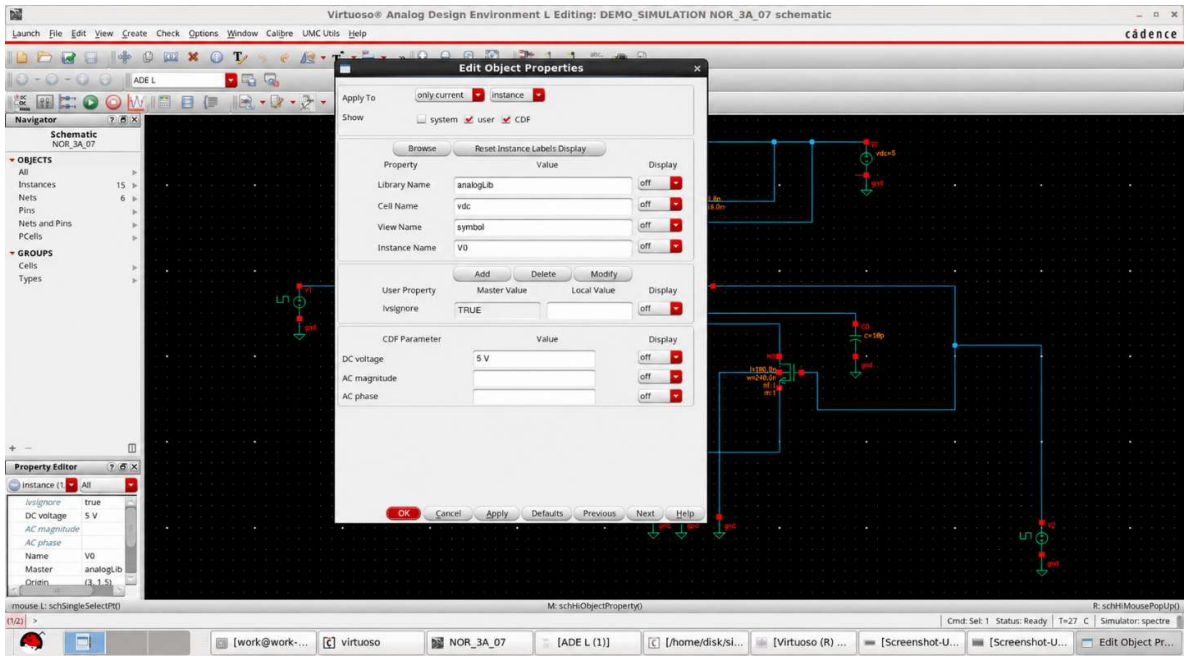
DESIGN AND SIMULATION SET-UP:

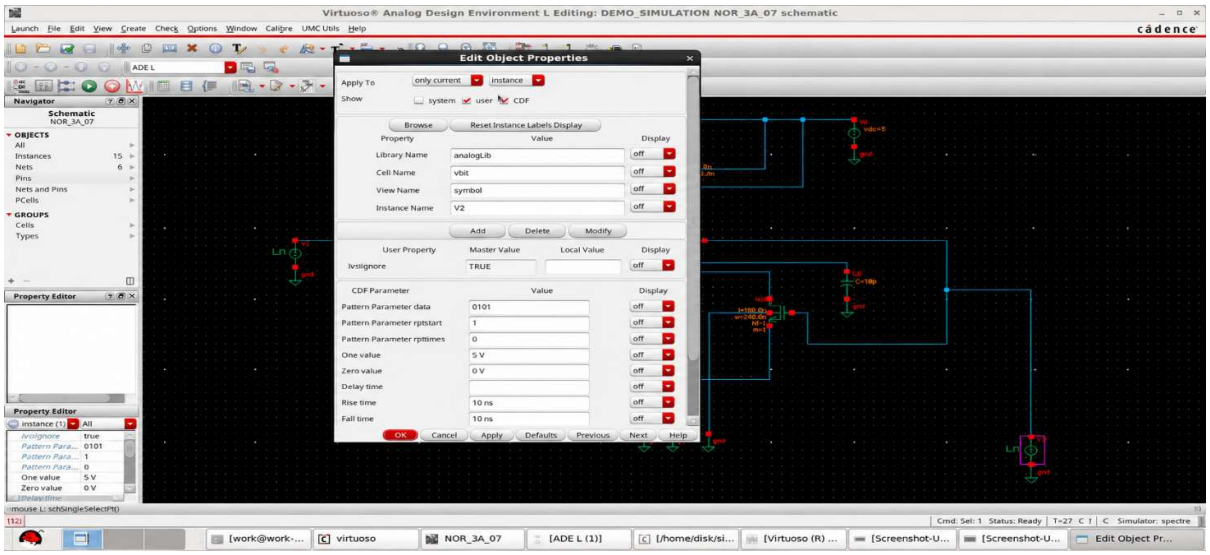
a) Design

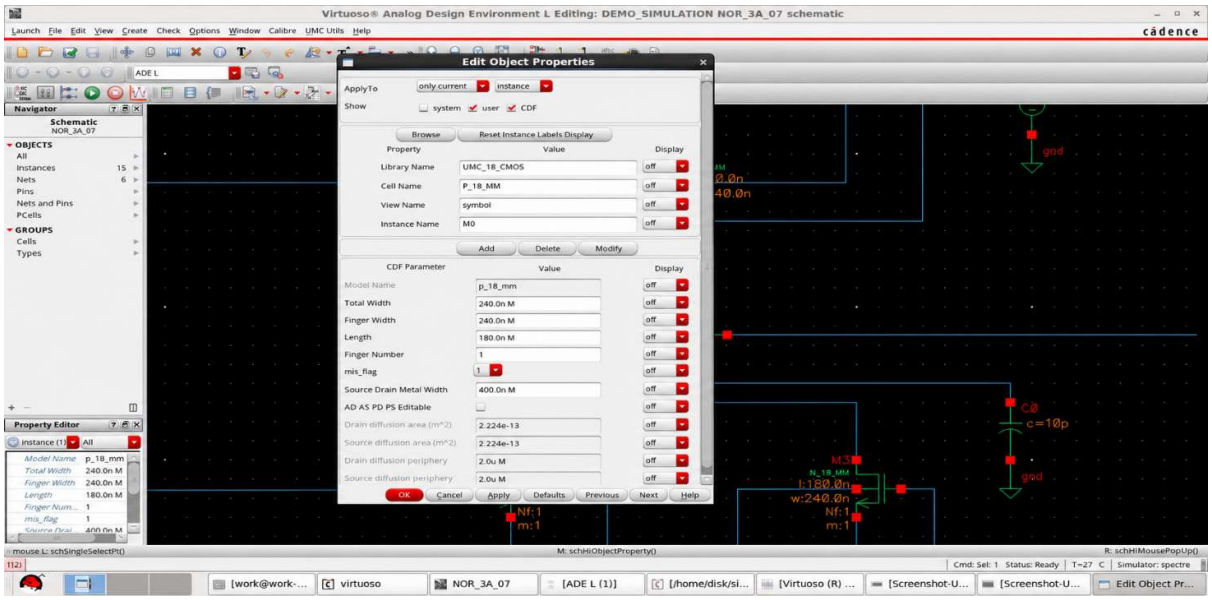



Virtuoso® Analog Design Environment L Editing: DEMO_SIMULATION NOR_3A_07 schematic
Edit Object Properties
Apply To
only current
instance
Show
system
user
CDF
Browse
Reset Instance Labels Display
Property
Value
Display
Library Name
analogLib
Cell Name
vbit
View Name
symbol
Instance Name
V1
Add
Delete
Modify
User Property
Master Value
Local Value
Display
ivsignore
TRUE
CDF Parameter
Value
Display
Pattern Parameter data
0011
Pattern Parameter rptstart
1
Pattern Parameter rpttimes
0
One value
5 V
Zero value
0 V
Delay time
Rise time
10 ns
Fall time
10 ns
OK
Cancel
Apply
Defaults
Previous
Next
Help

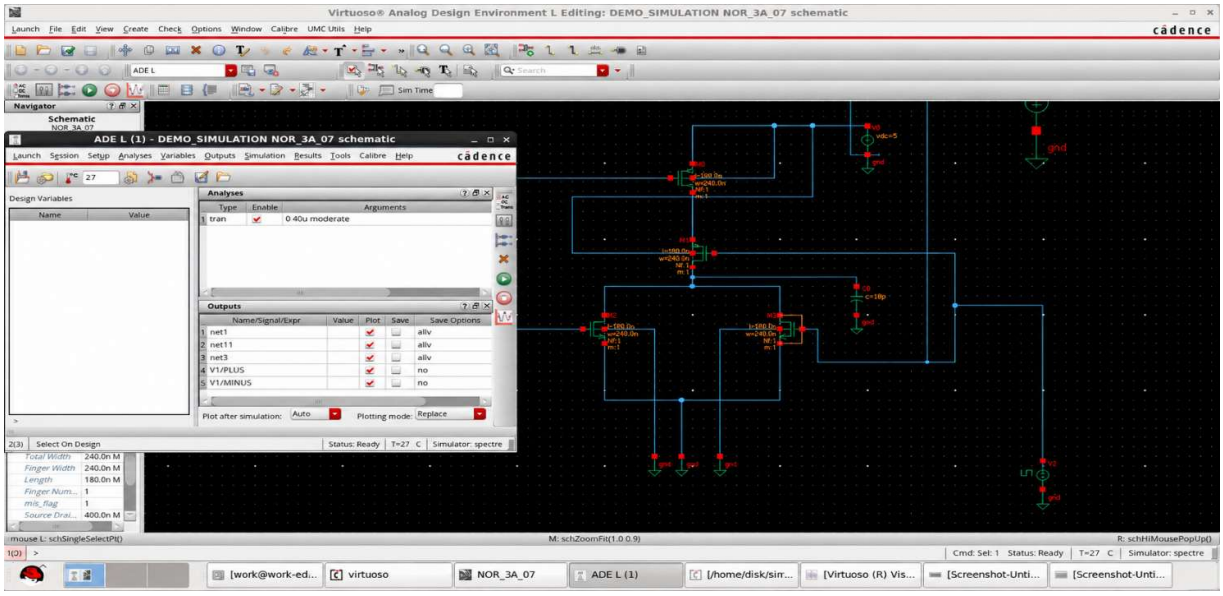

Virtuoso® Analog Design Environment L Editing: DEMO_SIMULATION NOR_3A_07 schematic
Edit Object Properties
Apply To
only current
instance
Show
system
user
CDF
Browse
Reset Instance Labels Display
Property
Value
Display
Library Name
analogLib
Cell Name
cap
View Name
symbol
Instance Name
C0
Add
Delete
Modify
CDF Parameter
Value
Display
Model name
Capacitance
10 p F
Width
Length
Multiplier
Scale factor
Initial condition
Temperature coefficient 1
Temperature coefficient 2
Temperature difference
OK
Cancel
Apply
Defaults
Previous
Next
Help


Virtuoso® Analog Design Environment L Editing: DEMO_SIMULATION NOR_3A_07 schematic
Edit Object Properties
Apply To
only current
instance
Show
system
user
CDF
Browse
Reset Instance Labels Display
Property
Value
Display
Library Name
analogLib
Cell Name
vdc
View Name
symbol
Instance Name
V0
Add
Delete
Modify
User Property
Master Value
Local Value
Display
ivsignore
TRUE
CDF Parameter
Value
Display
DC voltage
5 V
AC magnitude
AC phase
OK
Cancel
Apply
Defaults
Previous
Next
Help

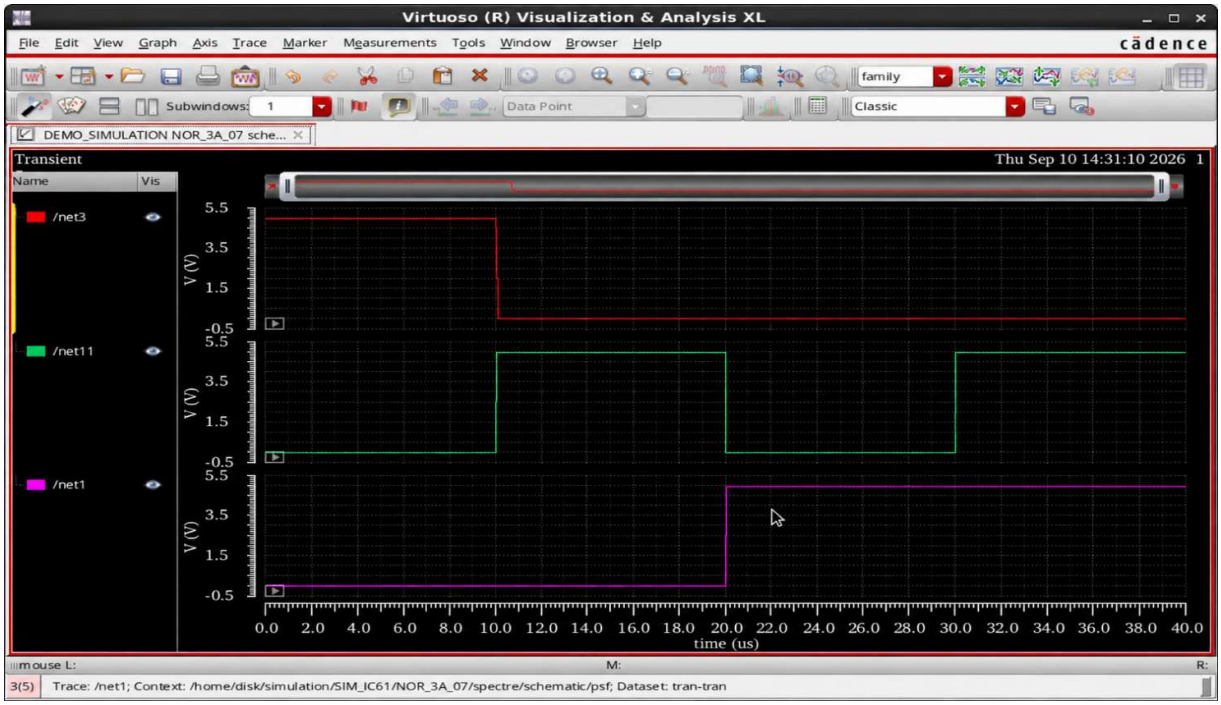




b) ADEL Simulation Setup



c) Transient Simulation Result



d) Conclusion

In this experiment, we have learnt the design of a CMOS NOR gate using the UMC 180 nm technology library (UMC_18_CMOS PDK) in a professional SPICE environment (Cadence Virtuoso). Subsequently, the CMOS circuit has been validated using transient simulation in the ADE-L environment. From this experiment we learnt how PMOS and NMOS transistors are connected in complementary pull-up and pull-down networks to realize the NOR logic function, how the devices are selected from the foundry library and placed in the schematic editor, and how the schematic is checked and saved before creating a symbol for it. We also learnt to build a test bench with the DC supply and multiple pulse inputs, to configure the transient analysis in ADE-L, and to plot the input and output waveforms together for verification.

The simulated output verifies the NOR operation of the circuit. The output is high only when all the inputs are low, and it becomes low whenever any one of the inputs is high. The output transitions follow the applied input signals correctly and maintain the expected logic levels throughout the simulation. This confirms the correctness of both the CMOS NOR schematic and the simulation setup, and establishes the fundamental design flow for implementing and validating CMOS combinational logic circuits in Cadence Virtuoso.

Signature of the Lab-in-Charge